BRAIN TUMOR CLASSIFICATION USING TRANSFER LEARNING

Dec 3, 2025

ABSTRACT

This project explored the application of transfer learning for automated brain tumor classification from MRI scans. ResNet50, a deep convolutional neural network pretrained on ImageNet, was used with a custom fully connected decoder to classify brain MRI images into four categories: glioma, meningioma, pituitary tumors, and healthy tissue. This classification mimics the clinical task in which MRI images are analyzed by doctors for the presence of tumors. Our approach combined the pretrained ResNet50 base with a custom classification head, using techniques like data augmentation, class weighting, and label smoothing to improve model performance.

1 Introduction

Brain tumors are serious medical conditions that require accurate diagnosis for effective treatment. MRI scans are commonly used to detect and classify brain tumors, but manual interpretation by radiologists is time-consuming and requires significant expertise. This creates an opportunity for automated classification systems that could assist medical professionals and potentially speed up the diagnostic process.

Deep learning, particularly Convolutional Neural Networks (CNNs), has shown great success in analyzing medical images (He et al., 2016). However, training deep neural networks from scratch typically requires large amounts of labeled data. This poses a problem considering the difficulty with acquiring medical data in some settings, and also the time/cost necessary in having a large set of these images labeled. Transfer learning offers a solution to this problem. Instead of training a model from scratch, one could use a model pretrained on a large dataset and adapt it to a specified task. A pretrained model would have ideally already learned to recognize basic visual features like edges, textures, and shapes, which are useful across many different types of images. This feature representation can then be fine-tuned or new layers can be added to specialize the model for a particular task.

For this project, we implemented a brain tumor classification model using ResNet50 (He et al., 2016), a popular deep learning architecture pretrained on ImageNet (Deng et al., 2009). Our goals were to build a classifier that could distinguish between four categories (glioma, meningioma, pituitary tumors, and healthy brain tissue), use transfer learning with ResNet50 to work effectively with a limited medical imaging dataset, and apply techniques like data augmentation and regularization to optimize the model. Through this project, we aimed to apply transfer learning and understand how pretrained models could be adapted for specialized tasks like medical image classification.

2 Background

ResNet, introduced by He et al. (2016), is a deep neural network architecture that uses skip connections to enable training of very deep networks. The key innovation is that skip connections allow gradients to flow more easily during training, solving the "vanishing gradient" problem that makes it difficult to train deep networks with many layers. ResNet50, which has 50 layers, has become widely used for image classification tasks and is a popular choice for transfer learning because it is pretrained on ImageNet (Deng et al., 2009), a large dataset of over 1 million natural images.

Transfer learning is especially useful in medical imaging because obtaining large labeled datasets poses a challenge. Research by Tajbakhsh et al. (2016) showed that models pretrained on ImageNet (Deng et al., 2009) can significantly improve performance on medical imaging tasks, even though natural images and medical images look quite different.

Tajbakhsh et al. evaluated CNNs with and without transfer learning through a multidiscipline (radiology, cardiology, and gastroenterology), multitask (classification, detection, and segmentation), and multimodal study. They found across all tasks, including classification tasks in radiology, that models with transfer learning improved performance, if not performing as well as traditional CNNs.

Transfer learning can improve results because the low-level features (like edges and textures) that the model learns are useful across different types of images. The main advantages of transfer learning are: the pretrained model has already learned useful visual features, less training data is needed because we're starting from a good initialization, and overfitting is mitigated by training images on the custom decoder layers only.

3 Methods

3.1 Dataset

A Brain Tumor MRI Dataset from Kaggle (Nickparvar, 2021) was used, which contained 7,023 MRI images divided into four classes: glioma, meningioma, pituitary tumor, and no tumor (healthy brain tissue).

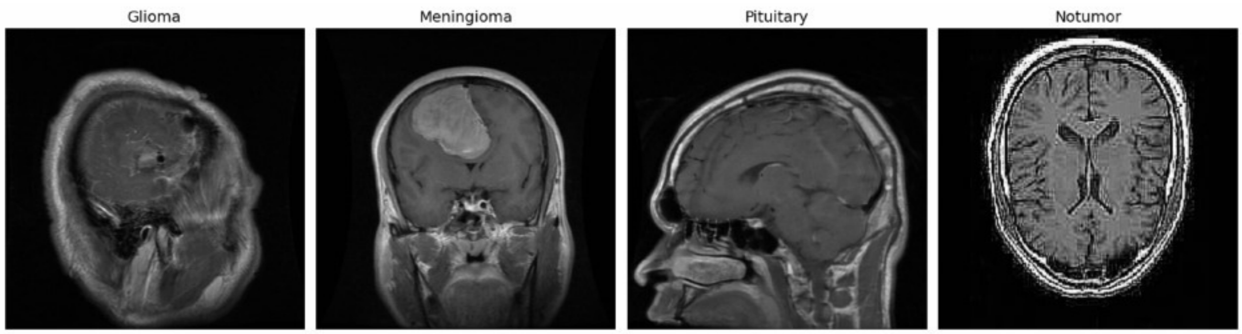

Figure 1: Sample images from different categories

The first three of these classes show brain tissue featuring a tumor of some kind. In some cases the tumors are more visually obvious, in other cases not. The dataset came pre-split into training and testing sets. We further split the training set, reserving 20% of this subset for validation, resulting in overall approximately 65% for training, 16% for validation, and 19% for testing. Figure 2 shows the distribution of data across classes and splits. Class imbalances were noted, as

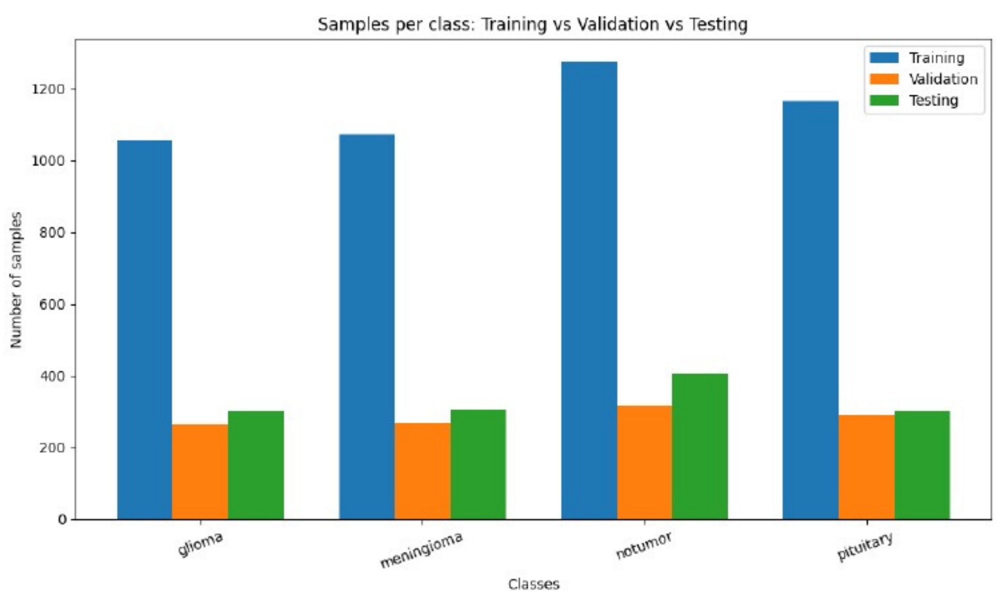

Figure 2: Distribution of brain tumor MRI samples across classes and data splits.

there were less samples for glioma and meningioma images. This was later handled by using class weights (Table 1).

3.2 Data Pre-processing and Augmentation

Since we used a ResNet50 model pretrained on ImageNet (Deng et al., 2009), images were pre-processed using ResNet-specific pre-processing, which converted RGB images to BGR format and zero-centered them with ImageNet statistics. All images were resized to 224×224 pixels with 3 color channels (BGR) to match what ResNet50 expected as input. To help our model generalize better and avoid overfitting, we applied data augmentation to the training set with the following parameters: rotation range of 20 degrees, width and height shift range of 0.2 (20%), horizontal flips enabled, zoom range of 0.2 (20%), and brightness range between 0.8 and 1.2. All samples from training, validation, and test sets had the same pre-processing applied. Augmentation was only applied to samples in the training set.

3.3 Model Architecture

The model used transfer learning with ResNet50 (He et al., 2016) pretrained on ImageNet (Deng et al., 2009). We started with the ResNet50 base model (removing the original top classification layer) and froze all the convolutional layers during training to keep the pretrained ImageNet features intact. After the ResNet50 base, we added a global average pooling layer to reduce the spatial dimensions down to a single feature vector. On top of that, we built a custom classification head consisting of three dense layers: a 512-unit layer with ReLU activation and 50% dropout, a 256-unit layer with ReLU activation and 40% dropout, and a 128-unit layer with ReLU activation and 30% dropout, followed by an output layer with 4 units and softmax activation for our 4 classes.

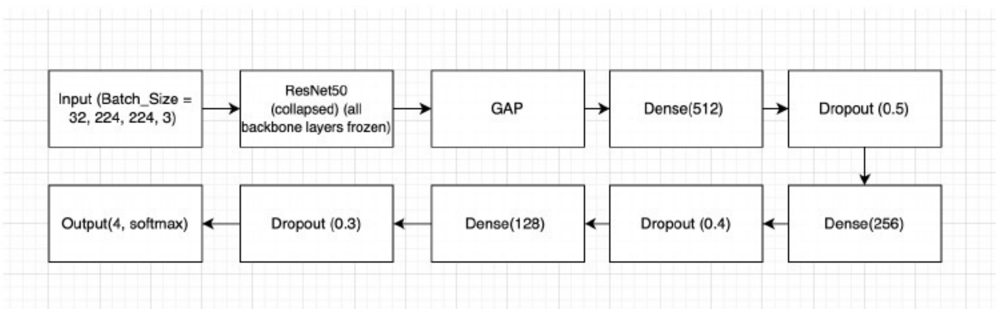

Figure 3: Architecture of the brain tumor detection model

Figure 3 illustrates the model, which in its entirety contained 24,801,540 parameters, of which 1,213,828 were trainable. This approach let us leverage the general visual features ResNet50 learned from ImageNet while training only the final layers to specialize in our dataset and task. By doing so the training and evaluation time for the model was also reduced several fold over a previous fully convolution model we designed, only taking approximately an hour to run rather than several.

3.4 Model Configuration

Label smoothing (0.1) was used with the categorical cross entropy loss to prevent the model from becoming overconfident and to help it generalize better. The learning rate, initially set to 0.0001, automatically decreased by half if the validation loss didn't improve for 5 consecutive epochs. Early stopping was also implemented - if the validation loss didn't improve for 10 epochs, training stopped and the best model would be restored (based on validation accuracy). In the case that the model did not use early stopping, the maximum epoch was set to 100, though in all iterations of the model, this was never met.

Table 1: Per-Class Accuracy

Class	Accuracy	Weight
Glioma	82.67%	1.08
Meningioma	76.47%	1.07
Pituitary	98.67%	0.98
No Tumor	97.28%	0.90

To handle the class imbalance in our dataset, we computed balanced class weights included in Table 1 using scikit-learn's compute_class_weight function with the 'balanced' strategy. This method calculated weights inversely proportional to class frequencies using the formula: $w_i = \frac{n_{samples}}{n_{classes} \times n_{samples,i}}$, where w_i is the weight for class i, $n_{samples}$ is the total number of training samples, $n_{classes}$ is the number of classes (4), and $n_{samples,i}$ is the number of samples in class i. These weights were applied during training to ensure the model paid appropriate attention to all classes regardless of their sample sizes.

4 Results

4.1 Baseline Model

A baseline convolutional model was generated before the application of transfer learning. This baseline model consisted of 3 convolutional blocks (each with Conv2D, BatchNormalization, MaxPooling, and Dropout), followed by global average pooling, two dense layers (256 and 128 units with dropout), and an output layer with 4 units and softmax activation. The baseline model used the same hyperparameters, data augmentation, and procedures as our transfer learning model. This model resulted in an overall test accuracy of 74.5%.

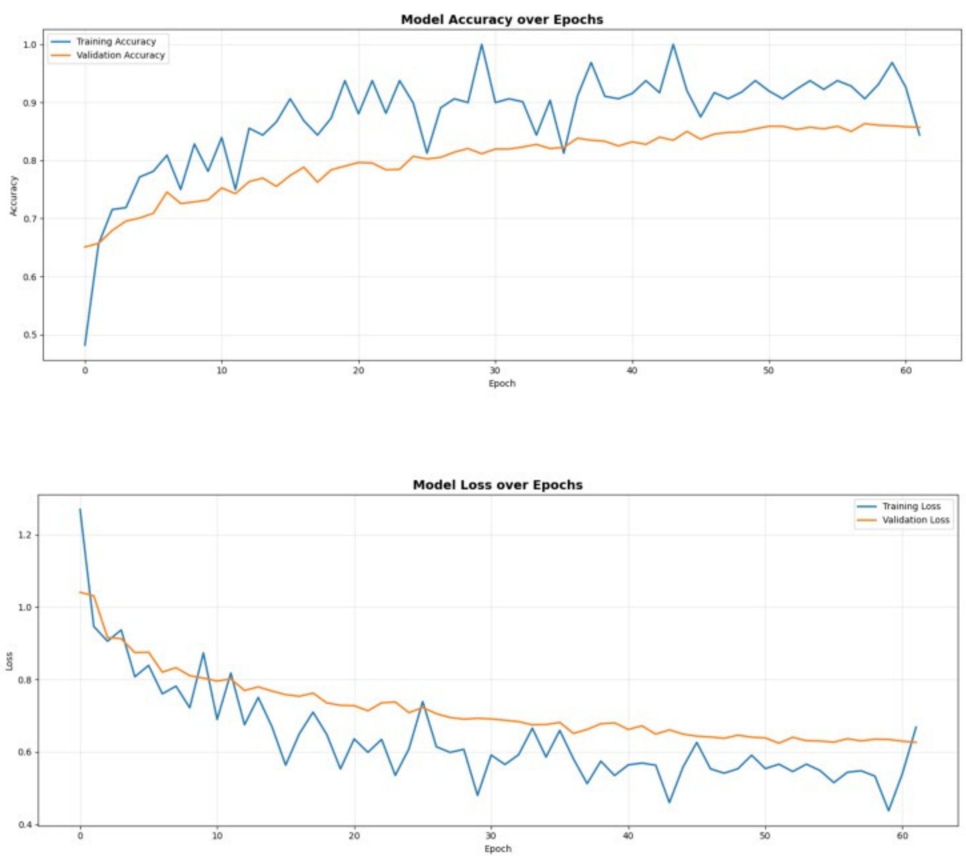

Figure 4: Training and validation accuracy and loss per epoch.

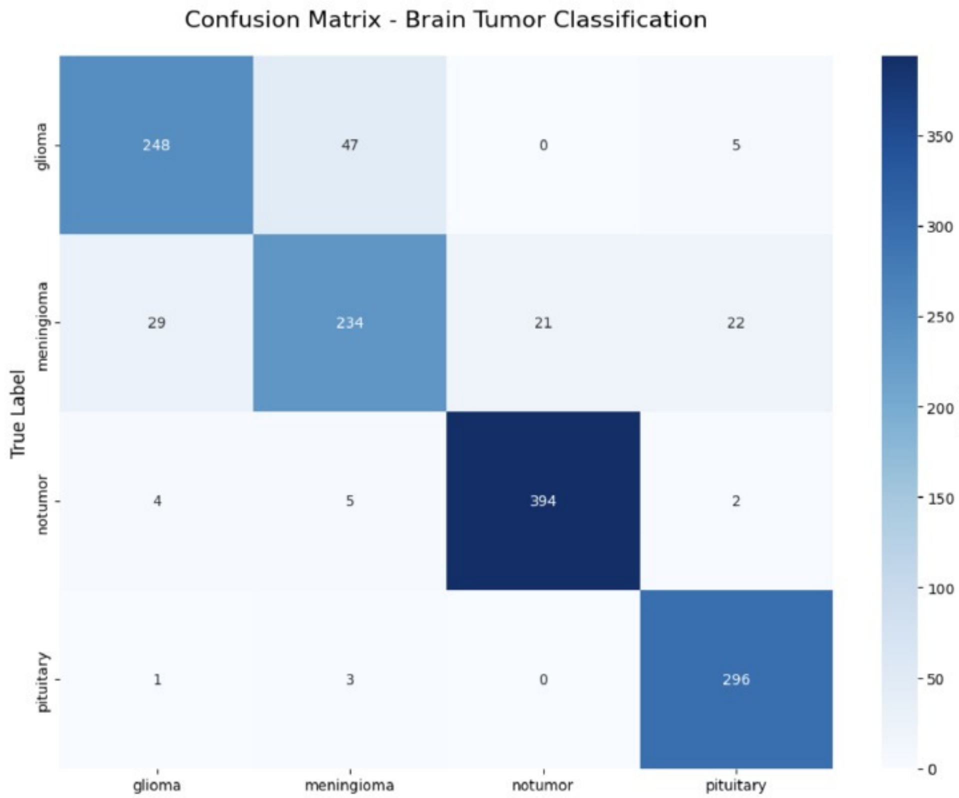

Figure 5: Confusion matrix for test set predictions showing per-class performance.

4.2 Transfer Learning Model

The transfer learning model trained successfully, reaching a training accuracy of approximately 84.38% and validation accuracy of 85.71% (Figure 4) in the last epoch, 62. This model resulted in a final test accuracy of 89.4%. The training curves showed good generalization - the validation metrics followed the training metrics closely without showing major signs of overfitting. It seems the methods to prevent overfitting - dropout layers, label smoothing, and data augmentation

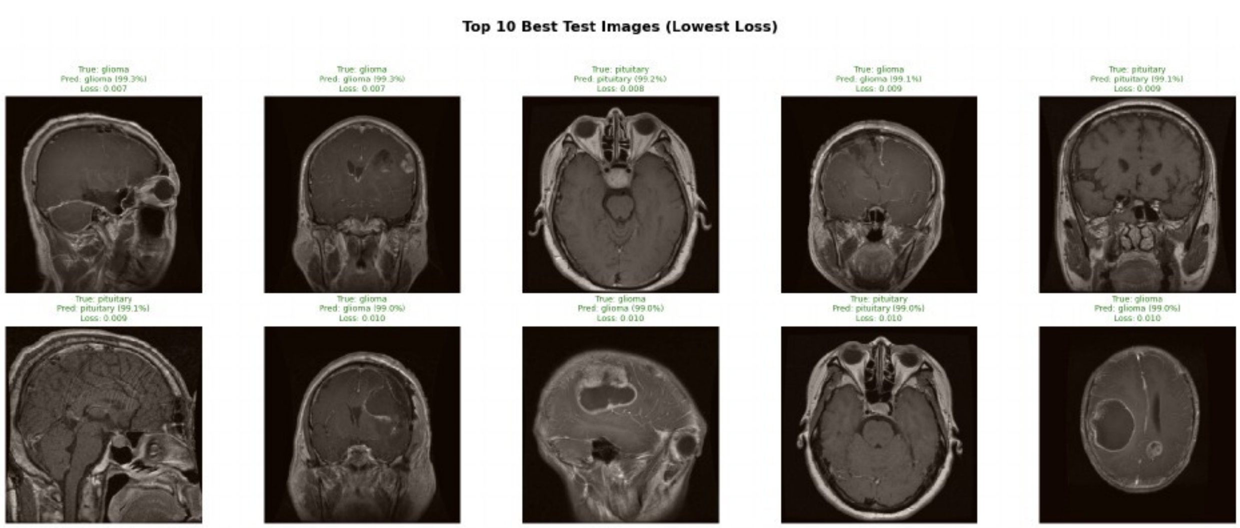

Figure 6: Top 10 images

- succeeded. Additionally, application of transfer learning and class weights improved the model's accuracy over the baseline model by 8% and 7%, respectively. The model performed well across all four classes (Figure 5) , with especially high accuracy on images with pituitary tumors, and lower accuracies on images with meningiomas and gliomas.

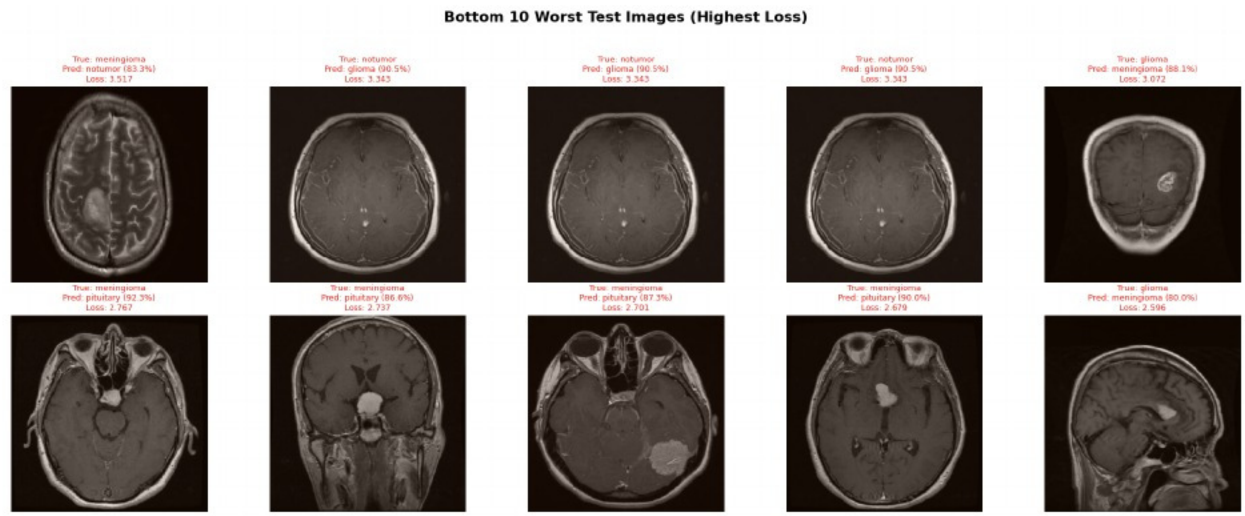

Figure 7: Bottom 10 images

As shown in Figure 6, the 10 best performing test images with the lowest loss mostly appear to have visually discernible tumors present, although some not. These MRI scans are from a variety of perspectives, and for the most part appear to have good image quality. The bottom 10 worst performing images in Figure 7 with the highest loss were mixed between images with and without obvious tumors. Several of the images mistake healthy tissue for having a glioma, and most others mistake one type of tumor for the other.

5 Discussion

5.1 Key Findings

Our results demonstrated that transfer learning with ResNet50 (He et al., 2016)worked much better than training a CNN from scratch for brain tumor classification. The pretrained ImageNet (Deng et al., 2009) features helped the model learn effectively regardless of the limited medical imaging dataset. The model achieved high accuracy across all four tumor classes, supporting the idea that visual features learned on natural images can transfer well to medical images. These results strongly align with other findings (Tajbakhsh et al., 2016) that support the application for transfer learning with ResNet50/ImageNet for classification tasks in radiology.

We believe transfer learning worked well for several reasons: (1) ResNet50's deep architecture captured features at different levels, from simple edges and textures to more complex patterns, (2) freezing the pretrained layers gave us a solid feature extractor without needing to train millions of parameters, and (3) our custom classification head was able to adapt those general features to the specific task of identifying various brain tumors along with healthy tissue.

5.2 Successes

Several of our design choices helped the model perform well. Using ResNet50 improved results over a fully convolutional network, despite the differences between natural images and medical MRI. The data augmentation (rotation, shifts, flips, zoom, brightness changes) also helped the model generalize better and avoid overfitting. By giving higher weights to classes with fewer samples, we ensured the model recognized all tumor types well, rather than focusing on the more common ones. Using dropout layers (30-50%), label smoothing, and early stopping all helped prevent overfitting and improved the model's ability to generalize to unseen data. Finally, automatically reducing the learning rate when training plateaued allowed the model make big learning steps at first and then fine-tune with smaller adjustments later, which made the model more suitable to changes in the gradient.

5.3 Limitations

Natural images from ImageNet are quite different from medical MRI scans in terms of visual characteristics, colors, and content, and while transfer learning still worked well, this domain gap might limit how benefit is taken from the pretrained features. For time purposes the transfer learning model was frozen, and the decoder was trained on the data, although the model may have performed better if after this all layers were fine-tuned and trained together on the training data. There is a risk of overfitting with this technique though.

Images with meningiomas had lower accuracy, which might have been because of class imbalances in the amount of training data, or image quality within that class. Hypothetically, the class imbalance contribution would have been mitigated by the use of class weights. This is supported by the increase in accuracy specifically contributed to by this optimization technique. Another contribution to the difference in performance across tumor classes could be attributed to by the nature of the tumors themselves. Pituitary tumors, given their localization to the pituitary gland tend to remain in the same area within the brain. Meningiomas originate in the meninges in the brain, which are the tissues that surround and support the brain. This means that meningiomas appear in various locations towards the outside of the brain tissue, and may have a large part of their mass obstructed due to imaging perspective. Gliomas, lastly, due to their origination from a particular type of neuron, tend to look very similar to neurons themselves, meaning they are difficult to distinguish from surrounding tissue. This corresponds with the findings in Figure 7 relating the different types of images that did not perform well.

5.4 Future Work

Beyond fine tuning, predictions from multiple different architectures (like ResNet, DenseNet, and EfficientNet) might make the model more robust and accurate. Adding attention layers or creating saliency maps would help us understand which parts of the images the model is focusing on, making it more interpretable. Training the model on related tasks at the same time (like tumor detection, segmentation, and grading) could help it learn better features as well. Finally, working with radiologists to evaluate the model's predictions and compare them to human performance would be important if this were to be used in a real clinical setting. It may also be worth investigating the implementation of a fully convolutional decoder, and other methods to split the data or stratify across groups.

References

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